

Lecture 07 — Fourier Series

MATH 2201 · Week 4

Any periodic function $f(x)$ with period $2L$ can be represented as an infinite sum of sines and cosines. This representation is called the Fourier series of f .

$$f(x) = a_0/2 + \text{SUM}[a_n \cos(n \pi x / L) + b_n \sin(n \pi x / L)]$$

The coefficients are obtained by exploiting orthogonality of the basis functions over one full period. Multiplying both sides by $\cos(m \pi x / L)$ and integrating term by term collapses the sum to a single surviving term.

$$a_n = (1/L) \text{INTEGRAL}[-L..L] f(x) \cos(n \pi x / L) dx$$

$$b_n = (1/L) \text{INTEGRAL}[-L..L] f(x) \sin(n \pi x / L) dx$$

Convergence is guaranteed at every point of continuity by the Dirichlet theorem. At a jump discontinuity the series converges to the midpoint of the jump, and the partial sums overshoot near the jump by roughly 9% of the gap. This is the Gibbs phenomenon and it does not vanish as more terms are added.